

PRESENTATION SCRIPT

Worth the Investment?

Capstone Event — Final Presentation Script

Tuesday, September 29, 2026 · Presented by Mahmoud

Script follows the "Worth_the_Investment_Capstone_Skeleton_Final.pptx" deck, slide by slide

How to use this script

- Each card below matches one slide, in order. The gold number is the slide number, so it is easy to check you are on the right one.
- The main paragraph is what to say out loud. It is written in plain, spoken language, not slide text, so it will not sound like you are reading bullet points.
- The gold “if asked” boxes are optional. Only use them if someone actually asks a follow-up question on that slide.
- You do not need to say every line word for word. Treat this as a safety net, not a script to recite.

17

slides

~10

min talk time

2

audience polls

Sep 29

capstone event

What this presentation needs to cover

This is the final capstone presentation: the full story, from the business question through the honest results to the live Tableau walkthrough and your recommendations.

The deck is built to be honest, not triumphant: two of the three hypotheses did not fully confirm, and the value forecast underperforms a coin flip. That is reported plainly on purpose. A jury will read a clear, well-defended null result as more credible than a forced win, so do not soften it live, lean into it.

OPENING LINE, BEFORE SLIDE 1

“Good morning, thank you for having me. My project is called Worth the Investment, and it asks a simple question: once you account for injury risk, can you actually trust a rising transfer value as a recruitment signal? I pulled together three real data sources, tested three hypotheses for three different stakeholders, and I'm going to walk you through what held up and what didn't, honestly. I'll finish with a live look at the Tableau dashboards, and I'm happy to take questions any time.”

THE SCRIPT · SLIDES 1 TO 17

1

TITLE SLIDE

Worth the Investment?

SAY

“This is our title slide. The project forecasts player value trend and injury risk, to support Premier League recruitment decisions. Three data sources, three stakeholders, and one honest recruitment signal, that's the whole project in one line.”

SAY

"Here's the plan. First, the business question, and why a rising value alone can be misleading. Then the data and the pipeline I built to combine three sources. Then three hypotheses, one for each stakeholder: scouting, front office, and medical staff. Then what the results actually mean, including where they didn't hold up. And at the end, I'll switch to Tableau and show the dashboards live."

SAY

"Here's the question the whole project is built around. My hypothesis was that market value trend alone may overstate a player's recruitment worth, because a player with rising value and high injury risk might not deliver the output that value implies. Quick definition before I go further: market value here means a player's estimated transfer-market worth from public valuation data, not a confirmed sale price. That splits into three testable questions: can we forecast a player's market value trend from his valuation history alone; do injury history, age, or position flag players with weaker value growth or higher injury risk; and combined, do value forecast and injury risk together beat value alone as a recruitment signal."

IF ASKED WHETHER THIS HYPOTHESIS WAS CONFIRMED

"Partly, and I'd rather say that up front than have it come out later. The injury-history half holds up well and is genuinely useful. The forecasting half didn't hold up, I'll show you both results honestly."

Three Sources, One Pipeline

SAY

"I combined three real sources: Transfermarkt for player values, Transfermarkt for injury records, and FBref for match performance across the top five leagues, 2017 to 2024. The pipeline is built with dbt and DuckDB, six models running end to end, from raw sources through staging and player-matching into one warehouse. In total, 1,959 players are in scope. As I'll get to later, only 649 of them have usable data from all three sources, that gap is real and I address it head-on rather than hide it."

IF ASKED ABOUT THE 1,959 VS. 649 GAP NOW

"1,959 is everyone in scope across the three sources. 649 have usable FBref playing-time data. I have a full slide dedicated to exactly this gap later, slide 12, so I'll go into the detail there rather than here."

Three Stakeholders, Three Hypotheses

SAY

"I built this for three different people who'd actually use it. Recruitment and scouting, who want to know if injury recency predicts value decline. Sporting directors and front office, who want to know if younger players have a better value-risk tradeoff. And medical and performance staff, who want to know if position predicts injury risk."

SAY

“Before I show you the answer, I want a quick show of hands. Which do you think predicts a player’s future value better: how recently he was last injured, or simply whether he has ever been injured at all?”

[Pause here for the show of hands, then click to reveal.]

“The answer is B. It's not how recent the injury is, it's whether he has any injury history at all. Players with no injury history grow 4.35% in value, in the median. Players with any injury history: flat, zero.”

SAY

“For scouting, my hypothesis was that injury recency predicts near-term value decline, that recent injuries hurt value more than old ones. That turned out to be mixed: injury history matters more than recency. Recent, moderate, and distant injuries all show the same flat value growth. The real split is having any injury history at all versus none, and this is based on almost 29,000 player-valuation snapshots, so it's not a small sample.”

IF ASKED ABOUT THE “+4.35% MEDIAN” NUMBER

“That's the median value change for players with no injury history, versus 0.0% for players with any injury history. I used the median rather than the average specifically because it's less distorted by a handful of huge outlier valuations.”

IF ASKED FOR THE STATISTICAL DETAIL BEHIND THAT NUMBER

“It's a one-sided Mann-Whitney test, significant at p approximately 0.0000 on close to 30,000 snapshots. The raw average gap is even bigger, +11.11%, but I report the more conservative +4.35% median because I excluded a handful of low-starting-valuation snapshots that looked like breakout-debut artifacts rather than a real injury effect.”

SAY

“For the front office, my hypothesis was that younger players show a better value-risk tradeoff than veterans. That’s a split result. Risk is confirmed: injury rate more than doubles with age, from about 2.5 to 6.0. But value tells a different story, only the 21 to 24 emerging band actually shows real value growth, about 7%. So it’s not simply that younger is always better, it’s specifically that one age band combining low risk with real upside.”

IF ASKED FOR THE STATISTICAL DETAIL

“Both differences are statistically significant on Kruskal-Wallis tests, n equals 582 players. Value growth: H equals 95.27, p approximately 0.0000. Injury rate by age: H equals 34.57, p approximately 0.0000. That’s one row per player, so this carries real weight.”

Position Doesn't Predict Risk, But Goalkeepers Stand Out

SAY

"For medical and performance staff, I hypothesized that position predicts injury and workload risk. Statistically, that's not confirmed, position alone doesn't predict injury risk with significance. But goalkeepers do trend lower, about 2.3 compared to roughly 4.2 for outfield players. I'm upfront that there are only 52 goalkeepers in this data, so I can't call it confirmed, but it's a real pattern worth flagging, not dismissing."

IF ASKED FOR THE STATISTICAL DETAIL

"Kruskal-Wallis across all four positions comes out to p equals 0.61, nowhere near significant. I'm reporting that as an honest null result rather than hiding it. The goalkeeper pattern lines up with real football knowledge, far less running and sprinting exposure, but with only 52 goalkeepers and a wide spread, it's underpowered to confirm, not disproven."

Guess the Number

SAY

"One more guess before the harder part. Backtested on 150 players, how often do you think my value forecast actually called the direction right, up or down?"

[Take one or two guesses from the room, then click to reveal.]

"The real number is 22.7%. That's worse than a coin flip, and I want to be upfront about that, not bury it."

The Forecast Is Worse Than a Coin Flip

SAY

"This is the honest result. I hypothesized that a player's market value trend could be forecast from historical valuation snapshots alone. It isn't reliable: backtested on 150 players, it gets the direction right only 22.7% of the time, well under the 50% you'd expect from a coin flip. And this isn't just random noise, it's confidently wrong in one direction: it predicted 'rising' 91% of the time, while 80% of the actual outcomes were falling. So there's a real, structural optimism bias in the model."

IF ASKED: "22.7% IS SO LOW THAT FLIPPING EVERY CALL WOULD SCORE ~77%"

"Exactly, and that's the point I'm making: this isn't randomly noisy, it's systematically biased toward predicting 'rising'. That's actually a more useful diagnosis than just 'the model is bad', it tells you precisely what's wrong with it."

IF ASKED: "A MODEL THAT ALWAYS SAID 'FALLING' WOULD SCORE ~80%"

"True, and worth having ready. But that's a constant-prediction shortcut that only works because most outcomes in this particular sample happened to fall, it wouldn't generalize. My model is at least trying to be directional, it's just biased, which is a specific, fixable kind of wrong. Either way, the conclusion stands: raw value-trend forecasting alone isn't trustworthy, which is exactly why I don't rely on it by itself."

IF ASKED ABOUT THE SIZE OF THE ERROR, NOT JUST THE DIRECTION

"Direction isn't the only problem, magnitude is worse. Median absolute error against the real held-out values is 149%, and it's worse for cheap or fringe players, about 184% median under one million euros, versus about 125% for players valued over ten million. So the model is least reliable exactly where scouting decisions are most marginal."

IF ASKED WHETHER THIS BIAS FEEDS INTO THE RECOMMENDATIONS DOWNSTREAM

"Yes, and I want to be upfront about that too. The recruitment-signal quadrants, Priority Target and Caution Flag, inherit the same 'rising' bias, about 93% of that combined table is flagged rising. So I treat that signal as a directional flag to investigate further, not a calibrated prediction to act on directly."

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THE HONEST CAVEAT

A Real Coverage Gap, 67% Excluded

SAY

"One more thing I want to be transparent about: data coverage. Of the 1,959 players in scope, 1,310, that's 67%, have zero recorded FBref playing time and had to be excluded from the position and age analyses. I checked this wasn't a matching bug: zero of those 1,310 have any exact FBref name match. Breaking that 1,310 down: 257 simply had no valuation yet inside FBref's 2017 to 2024 window, 269 more looked like possible matches but a manual review found only shared surnames, not real matches, and the remaining 784 had no FBref minutes in that window, or debuted after it closed. That leaves 649 players with usable data, and the age and position charts you saw earlier use an even stricter cut, n equals 582, after filtering for a reliable minimum sample per bucket."

IF ASKED WHY 1,959 / 649 / 582 ARE ALL DIFFERENT NUMBERS

"1,959 is everyone in scope across all three sources. 649 of those have any usable FBref playing-time data. The age and position breakdowns specifically use 582, a stricter subset after filtering out players without enough data to trust that bucket. Same population, progressively filtered for what each analysis actually needs."

SAY

“So, is it worth the investment? The clearest finding is that any injury history flattens value growth, that's the signal that actually holds up, and it's strong: almost 29,000 snapshots, highly significant. For front office, 21 to 24 is the true value-risk sweet spot. For medical, position risk isn't confirmed yet, but it's worth continuing to monitor.”

“And to directly answer the third question I raised at the start, does the combined signal beat value alone: not yet. Injury history is the reliable signal here, a rising valuation, or forecasting one, is not. So the combined signal doesn't beat value alone yet, because the forecast half doesn't hold up on its own.”

IF ASKED WHY BUILD A PROJECT WHERE TWO OF THREE HYPOTHESES DIDN'T CONFIRM

“Because that's what a rigorous test looks like, not every hypothesis is supposed to survive contact with real data, and reporting a null result honestly is exactly the skill this project demonstrates. The injury-history finding alone is a real, statistically strong, actionable signal, and I found it precisely because I tested rigorously instead of only keeping the results that looked good.”

SAY

“So here's what I'd actually recommend doing with this. Scouting: screen for any injury history, not just recent incidents, that's the pattern the data actually supports. Front office: weight the 21 to 24 age band in valuation models, it's the only bracket combining above-average growth with below-average risk. Medical and performance: keep monitoring workload regardless of position, the goalkeeper pattern is promising but not yet statistically confirmed.”

SAY

"Now I want to show you this live."

[Alt-tab to the Tableau window if it's already open, rather than clicking the button on the slide, that avoids opening a duplicate window. Only use the button if Tableau isn't open yet.]

"I'll start on the cover, then walk through Value Trend, Risk Summary, Age and Risk Profile, and Recency and Position Risk Profile, all cross-filterable through a shared set of parameters."

IF THE CONNECTION DROPS OR THE FILE WON'T OPEN

Note: the old speaker notes on this slide mention a static fallback screenshot on 'the very next slide'. After the reorder, the next slide is now Thank You, there is no fallback slide in the current deck. If Tableau fails to open live, narrate the same four views from memory instead (Value Trend, Risk Summary, Age and Risk Profile, Recency and Position Risk Profile) rather than expecting a rescue slide. Worth testing Tableau opens cleanly before you start tomorrow.

SAY

"That's the live walkthrough. To close: this project demonstrates a full three-source pipeline, honest hypothesis testing, one real and actionable signal, and live interactive dashboards, all built and presented end to end by me. The GitHub link on screen has the full repo if anyone wants to dig into the code. Thank you, I'm happy to take questions."

SAY

"I also have an appendix slide on the project roadmap and timeline, happy to pull it up if anyone wants to see how the four weeks broke down."

This slide is not part of the main flow, it sits after Thank You on purpose as backup material. Only bring it up if someone specifically asks about your process or timeline.

WRAPPING UP

MOVING INTO Q&A

After slide 16 (or 17 if you pulled up the appendix), pause, take a breath, and let the room lead. It's fine to say "let me check my notes" before answering something specific, that reads as careful, not unprepared.

If a juror pushes on the overall framing, whether a project with two null hypotheses out of three is still a strong capstone, the honest answer is yes: the injury-history signal is real, statistically strong, and actionable, and reporting the forecast's failure plainly, instead of hiding or overstating it, is itself the skill being assessed. Say that directly rather than getting defensive about it.